NOTE 2026-10-21 (owner rule D11, spend nothing): the paid parts of this report (Anthropic API, paid Gemini, any credit or deposit) are DROPPED. Free-only plan: docs/11 phase 7 and prompt Z1 in docs/12. The retrieval design, quote-verification algorithm, evaluation set, injection rules and cost-guard ideas below still apply.

Research 2026-10-21: ask-the-docs (cheapest safe build) — feeds phase 7

(1P) = first-party page fetched. Everything else third-party or UNVERIFIED. The fetch tool summarises pages with a small model, so confirm any number you rely on on the page itself.

Bottom line

Primary: Anthropic API, Claude Haiku 5.5 with the Citations feature, prepaid with a spend cap, about $0.0006 per question (~$0.60 per 1,000). Fallback: Gemini API paid tier (prepay $5 minimum), Flash-Lite, quote-then-verify JSON. Retrieval: Postgres full-text search by headings; add pgvector only if the 40-question test shows weak recall. Do NOT rotate five free OpenRouter accounts (breaks terms, pointless). Do NOT send private docs to a free tier that trains on data (conflicts with D1). NotebookLM (now "Gemini Notebook") cannot be called from the app: personal reader only.

Providers (as of 2026-10-21)

- Anthropic (1P platform.claude.com/docs/en/about-claude/pricing, /api/rate-limits, /build-with-claude/citations): Haiku 5.5 $0.10 in / $0.50 out per MTok up to 100k-token prompts, $0.50/$2.50 above; cache reads $0.01/MTok; Haiku 4.5 $1/$5; Sonnet 5.5 $2/$10; start tier $500/month cap, you can set lower; no free tier; inputs/outputs not used for training by default (privacy.claude.com/en/articles/7996868); citations work on all active models, ZDR-eligible, `cited_text` not counted as output; citations and structured outputs cannot be combined (HTTP 400). Minimum deposit UNVERIFIED.
- Gemini API (1P ai.google.dev/gemini-api/docs/{rate-limits (2026-10-15), pricing (2026-10-19), billing, terms}): free-tier numbers not published (AI Studio shows per project); free tier content may be used to improve products and read by humans; paid not used for training; EEA users require paid services; 3.1 Flash-Lite $0.25/$1.50, 3.5 Flash-Lite $0.30/$2.50 per MTok; prepay minimum $5.
- OpenRouter (1P openrouter.ai/docs/api-reference/limits, /terms): `:free` models 20 req/min and 50/day under $10 of credit, 1,000/day at $10+; extra accounts/keys do not change limits; terms 7(3) bans multiple accounts to circumvent limits; 6.1 some models may train on inputs. -> one account with $10 at most.
- Groq: free-plan figures unclear (gpt-oss-120b 30 req/min, 1K/day, 200K tokens/day, per organization). Cerebras: free trial 5 req/min, 1M tokens/day, "no permanently free tier". Mistral: official limits not found (404). Cloudflare Workers AI: 10,000 neurons/day free, bge-small embeddings $0.02/M tokens (1P, 2026-10-20). Hugging Face: free users no credits. GitHub Models: retired 2026-07-30 (1P). OpenAI: gpt-5-nano $0.05/$0.40 per MTok, text-embedding-3-small $0.02/MTok, no quote API.
- plan file docs/06-API-PLAN.md still says "5 keys rotation": remove it.

Retrieval

A few hundred pages ≈ 0.3-0.6M tokens: do not put everything in the prompt (above 100k tokens Haiku 5.5 costs 5x per token). FTS: stored generated tsvector, GIN index, websearch_to_tsquery, ts_rank in an RPC; heading weight A, body B (1P supabase.com/docs/guides/database/full-text-search). Chunk by h2/h3, keep heading path, split chunks

over ~600 tokens at paragraph breaks, store page id and character offsets; top 6 chunks. Urdu: probably the `simple` config (no Urdu stemmer) — from general knowledge, test it. Embeddings only if recall@6 < ~90%: Supabase hybrid search with RRF (1P supabase.com/docs/guides/ai/hybrid-search), gte-small 384 dims in an Edge Function (docs from 2024, recheck), Gemini Embedding 2 $0.20/M tokens, OpenAI 3-small $0.02/M; indexing 0.5M tokens costs $0.01-0.10. Gemini File Search tool (managed, 1 GB free store, Markdown supported) ties us to Google with coarse citations: skip.

Verbatim-quote verification (run on every answer)

Option A (primary): send each retrieved chunk as a `document` block with `citations: {enabled: true}`; API returns `cited_text` + location indexes. Option B (any model): JSON `{claims:[{text, chunk_id, quote}]}` or `{refuse:true}`. 1 Normalise quote and chunk: Unicode NFC, collapse whitespace, unify curly/straight quotes and dashes, strip Markdown emphasis markers; never strip Urdu characters. 2 Pass only if the normalised quote is a contiguous substring of the named chunk, and the chunk was one of those retrieved. 3 Reject quotes under 6 or over 80 words. 4 Display text from our database using the matched offsets, never the model's copy; link to the heading anchor. 5 If any claim fails retry once naming the failing quote; then show "No verified answer" with retrieved headings as links. 6 Substring proves existence, not support: use the golden set and an optional judge pass.

Evaluation (~$0.03 per run of 40 on Haiku 5.5)

24 single-page facts (8 with numbers/exact strings), 6 two-page questions, 4 paraphrases with little keyword overlap, 6 unanswerable, 2 targeting a planted injection page; hand-written gold page and gold quote each. Metrics: recall@6; quote validity 100%; answer correctness ≥ 90% (owner judged); groundedness (LLM judge "does the quote entail the claim?" + human check of 10); correct refusals 6/6; false refusals ≤ 10%. Re-run after every model, prompt or chunking change; keep history. (RAG Triad per OWASP LLM01:2025.)

Prompt injection

Documents are attacker-controlled (collaborators, AI tools). The Q&A function has no tools (one model call, no web, no writes, no MCP; text only). Never render model output as HTML/Markdown links or images; show plain text and database quotes; link only to our heading anchors. Delimit retrieved text in tagged blocks and say it is data. Retrieve under the caller's identity with RLS. Reuse the phase 6.7 scanner; keep injection items in the golden set. Residual risk: a planted real quote still passes substring: show the source page next to every answer.

Keys and cost guard (1P supabase.com/docs/guides/functions/secrets)

Key only as an Edge Function secret (dashboard or `supabase secrets set`, `Deno.env.get`); names cannot start with `SUPABASE_`; one key for this app in its own provider workspace with its own cap. Never in the bundle (`VITE_`), git, chat, plain DB column, localStorage, MCP token path, logs. Free plan: 500,000 function calls/month, 2 s CPU, 150 s wall clock. Keep JWT verification on; re-check membership in SQL; table `ai_usage(user_id, day, requests, est_cost_micro)` with atomic upsert RPC; ~30 questions/day per user, ~300/day global, kill switch; cap question 500 chars, context ~6k tokens, max_tokens ~600; cache by hash(normalised question, project, content version); hard monthly cap in the provider console (e.g. $5; billing data can lag ~10 min); browser never chooses model or provider.

Gemini app side

NotebookLM / Gemini Notebook: free 50 sources per notebook, each ≤ 500,000 words or 200 MB (1P support.google.com/notebooklm/answer/16215270); compute-based limits (AI Pro 4x, refresh 5 h, weekly cap) (1P support.google.com/gemininotebook/answer/17670842); read-only sharing; no MCP/API mentioned; merge several hundred pages into ≤ 50 files. Gemini app custom MCP app: see docs/14. Gems are removed from personal accounts from November 2026, replaced by skills (1P support.google.com/gemini/answer/15146780): do not build on Gems.

Recommended plan and monthly cost

7.1 answer Q8: no rotation, no free tier for private docs; deposit the minimum on Anthropic, cap $5, own workspace. 7.2 key via Supabase secret dialog. 7.3 FTS chunks + one Edge Function (auth, rate limit, retrieve, Haiku 5.5 with Citations, substring check). 7.4 build the 40-question set first. Later: Gemini paid fallback behind the same function; pgvector only on recall failure. Cost: 1,000 questions ≈ $0.60; 5,000 ≈ $3; total ≈ $1-3/month + one-time ~$5 deposit; with 100k-token contexts ~10x more (the 6k cap prevents it). Risks: a real quote can still mislead or be planted; price jump above 100k-token prompts; free-tier numbers unstable (Gemini, Mistral, Groq UNVERIFIED); free-tier privacy; Gemini custom apps US/English only; EEA needs paid Gemini; model names "Haiku 5.5/Sonnet 5.5" come from the live pricing page, no model was called (latency and citation quality untested).